Varsini Sakthivadivel Ramasamy

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EDUCATION

Johns Hopkins University – Baltimore, MD *May 2026*
Master of Science in Bioinformatics (3.9/4.0)
Thesis: scRNA-seq Representation Learning for Cell-Type Classification

VIT University – Vellore, India *May 2024*
Bachelor of Technology in Biotechnology (3.9/4.0)

RESEARCH EXPERIENCE

Research Assistant, *Johns Hopkins University School of Medicine*, Baltimore, MD *Jan. 2026 – Present*
PI: Dr. Maide Ozen

Project: Transformomics: Learned attention uncovers tissue boundaries in spatial transcriptomics cell state classification

- Independently developed hypothesis and structured project to test if learned spatial attention over local tissue neighborhoods improved interpretability of cell-state classification in spatial transcriptomics, relative to standard non-spatial baselines.
- Designed a transformer encoding each Visium spot jointly with its 6 nearest spatial neighbors, summing expression, coordinate, and token-type embeddings so attention weights over neighbors are learned.
- Validated on 10x mouse brain data (2,796 Visium spots, 2,000 HVGs), which showed via attention analysis that the model allocates disproportionate weight (54%) to spatial neighbors at anatomical boundaries vs. 2% in homogeneous tissue, and validated gene-saliency maps against canonical cell-type markers (Mbp, Gfap, Pcp4).
- Achieved 89% validation accuracy; Logistic-regression baseline outperforms on raw accuracy (92%), but the model's learned attention provides interpretable, biologically grounded insight that the baseline cannot offer.
- Deployed for public inference on HuggingFace Spaces: huggingface.co/spaces/Varsini-Sakthi/Transformomics

Project: Homeostatic, phagocytic, NRF2/Hmox1 oxidative, Apoc1 and chemokine microglia transcriptional programmes in naïve mouse cerebral cortex from embryo to adulthood

- Characterized microglial transcriptional heterogeneity across 5 developmental stages using pseudobulk RNA-seq pipeline integrating gene module scoring, PCA, differential gene expression, and pseudotime trajectory inference.
- Identified 4 distinct transcriptional programmes with coordinately shifting activity across development, revealing stage-specific regulatory transitions relevant to neuroinflammation and brain maturation,
- Results establish a developmental baseline for distinguishing normal microglial maturation from pathological activation in neurodevelopmental disease models.
- Writing first-author manuscript in preparation and developing tables and figures for publication.

Project: NRF2/HO-1 Antioxidant Pathway Dysregulation in Rodent Models of Perinatal Brain Injury: Rat Microglia and Mouse Oligodendrocyte Precursor Transcriptomes [Working Title]

- Analyzed bulk RNA-seq (rat microglia/macrophage) and microarray (mouse O4+ oligodendrocyte precursor cells) datasets to characterize NRF2/HO-1 antioxidant pathway dysregulation in rodent models of perinatal brain injury.
- Built domain-split protein-protein interaction networks (STRINGdb) and assessed cross-branch concordance to identify shared regulatory nodes.
- Identified that O4+ oligodendrocyte precursor cells mount a more diagnostically separable transcriptional response to IL-1β than microglia, and nominating OPCs as an additional vulnerable cell population in perinatal brain injury.

Project: Translational Validation of NRF2/HO-1 Antioxidant Signaling in Human Perinatal Brain Injury: A Cross-Cohort Transcriptomic Analysis [Working Title]

- Validated rodent-derived NRF2/HO-1 pathway findings in human perinatal brain injury by integrating transcriptomic data across 4 independent human cohorts.
- Applied timepoint-adjusted limma differential expression and pseudobulk clustering to assess translational concordance between rodent and human NRF2/HO-1 pathway signatures.

- Results showed the MHC-I/interferon/chemokine-NRF2/Hmox1 signature identified in rodents was recoverable across independent human cohorts once developmental timepoint was accounted for, extending the rodent-model findings toward clinical relevance.

Master's Thesis, *Johns Hopkins University, Department of Bioinformatics,* Baltimore, MD *Dec. 2025 – Apr. 2026*

Project: scRNA-seq Representation Learning for Cell-Type Classification

- **Developed** the hypothesis that a nonlinear VAE embedding (scVI) would outperform linear dimensionality reduction (PCA) for cell-type classification, and designed a controlled benchmarking framework to isolate the effect of the embedding method.
- Trained a 30-D scVI variational autoencoder in Python against a PCA baseline on 12,012 PBMC cells (36,601 genes, 2,000 HVGs).
- Annotated 12 immune cell populations via Leiden clustering and marker-gene differential expression (Wilcoxon test).
- Results showed PCA outperformed scVI consistently, indicating that added model complexity did not translate to better downstream performance.

Research Assistant, *Johns Hopkins University,* Baltimore, MD *Jun. 2025 – Mar. 2026*

Project: Multiclass Brain Tumor Classification from MRI Radiomics

PI: Dr. Sajung Yun

- Built a multiclass brain tumor classification pipeline on 3,000+ MRI slices.
- Achieved 97% accuracy with XGBoost; applied SHAP to show GLCM texture descriptors of tumor heterogeneity drove classification, aligning with established radiological malignancy markers.
- Reported macro-averaged precision/recall (0.59/0.63) alongside accuracy to surface reduced performance on minority tumor classes leading to the next phase of clinically urgent subtype classification.

Summer Intern, *Johns Hopkins University School of Medicine,* Baltimore, MD *May 2025 – Sep. 2025*

Project: Immune Dysregulation in a Murine Cystic Fibrosis Model

PI: Dr. Dao-Fu Dai

- Quantified immune dysregulation in a murine cystic fibrosis model via qPCR, ELISA, flow cytometry, and confocal immunofluorescence imaging of CD45+, CD3+, F4/80+, and Ly6G+ marker populations.

PUBLICATIONS

Published

- Antony A., **Sakthivadivel Ramasamy V.,** Veerappapillai S., Karuppasamy R. (2025). "Genome mining of secondary metabolites from *Streptomyces spp*. for the screening of novel and potent fungicide: Focus on rice blast disease." Research Journal of Chemistry and Environment 29(10):66–75.
- **Sakthivadivel Ramasamy V.,** Bhor S.S. (2023). "An Integrated Study to Extrapolate the Interaction of NMDAR with Potential Ligands for the Treatment of Alzheimer's Disease Symptoms." International Journal of Biochemistry and Biomolecule Research 1(1):15–27.

Manuscripts in Preparation

- **Sakthivadivel Ramasamy V**., Ozen M. "Homeostatic, phagocytic, NRF2/Hmox1 oxidative, Apoc1 and chemokine microglia transcriptional programmes in naïve mouse cerebral cortex from embryo to adulthood." Manuscript in review.
- **Sakthivadivel Ramasamy V.,** Ozen M. "Developmental regulation of the *Hmox-1* regulatory network in microglia and oligodendrocyte progenitor cells during white matter injury: Cell specific inflammatory responses." [Working Title]
- **Sakthivadivel Ramasamy V.,** Ozen M. "Translational Validation of NRF2/HO-1 in Human PBMCs from extreme preterm neonates to adulthood." [Working Title]
- **Sakthivadivel Ramasamy V.,** Ozen M. "Transformomics: Learned attention uncovers tissue boundaries in spatial transcriptomics cell state classification." [Working Title]

TEACHING EXPERIENCE

Teaching Assistant, Johns Hopkins University, Bioinformatics *Aug. 2025 – May 2026*

- Supported instruction of Python, UNIX, and SQL for a graduate-level bioinformatics computing course, leading lab sessions and troubleshooting student code and pipeline errors.

- Built automated RNA-seq and proteomics analysis pipelines in Python and R for in-class demonstrations and as reusable templates for student workflows, translating research-grade pipeline code into accessible teaching material.

PROFESSIONAL EXPERIENCE

AI Trainer – Micro1 *Aug. 2026 – Present*

- Evaluate AI assistants on real-world email and follow-up workflows on mobile, approving or stopping the agent at key steps, screen-recording each run, and scoring completion, quality, control, satisfaction, and efficiency against a rubric.
- Ran head-to-head evaluations of Claude, Muse, Grok, Perplexity, and M365 Copilot on multi-app business workflows across Snowflake, Google Drive, Gmail, and Google Calendar, using identical prompts on every platform.
- Seeded Snowflake and Google Workspace test environments with known ground-truth answers, including metrics requiring multi-table joins that naive queries get wrong, and reset them between runs so every platform started from an identical state.
- Verified what each agent actually did in the apps against expected end states (calculated metrics, draft emails, calendar invites) instead of trusting the agent's own summary, flagging join errors and distribution-rule violations.
- Documented failure modes, disclosure accuracy, and safe-write boundary violations (e.g., sending vs. drafting) in written evaluations, and participated in peer calibration sessions to keep scoring consistent.

Data Analyst Intern – INGEST Project, Johns Hopkins University *Oct. 2024 – Apr. 2025*

- Built a computer vision data quality assurance pipeline (CVAT, 3DSRBench, Python) to validate infant keypoint annotations for a developmental research study, applying inter-rater reliability metrics to flag discrepancies.

Project Intern – BioAxis DNA Research Centre *Jan. 2024 – Apr. 2024*

- Investigated amyloid and tau pathology in Alzheimer's disease through genomics, proteomics, and structural biology analysis, applying immunoinformatics and SNP/alternative splicing analysis to characterize proteins retrieved and cross-referenced public expression datasets (GEO) and genomic annotations (UCSC Genome Browser) to contextualize candidate genes and regulatory regions implicated in neurodegeneration.
- Analyzed drug development strategies for Transthyretin Amyloidosis, examining candidate molecular targets and therapeutic options relevant to protein misfolding disease.
- Explored vaccine optimization for *Staphylococcus aureus* using computational structural biology tools (RasMol, PVS, EMBOSS, ArgusLab), including antigenic prediction to identify candidate epitopes for vaccine design.

Intern – BioNome, Genomics & Bioinformatics Solutions *Jan. 2023 – Apr. 2023*

- Applied molecular docking (PyRx, AutoDock), MD simulations (GROMACS), QSAR modeling, and ADMET/pharmacophore screening to evaluate ligand-target binding for Alzheimer's disease drug candidates.
- Modeled and validated target protein structures using homology modeling (MODELLER) and structure-prediction tools (AlphaFold), assessing model quality with ProSA-web and ExPASy prior to docking.
- Performed SNP calling from whole-exome sequencing data using Galaxy-based NGS workflows and subtractive proteomics for hypothetical protein annotation,
- Improved identification of novel disease-relevant targets and potential drug candidates from previously uncharacterized genomic regions.
- **Wrote a first-author journal article**; prepared tables and figures, and managed the paper publication process.

SKILLS

Machine Learning: Supervised/unsupervised learning, representation learning, variational autoencoders (VAE), XGBoost, Deep learning (PyTorch), hyperparameter tuning, LLMs, model evaluation (ROC-AUC), feature engineering

Programming: Python, R, SQL, Snowflake, Bash, Linux/Unix, MATLAB, SAS, AWS, Jupyter, Git, C++

Bioinformatics & Genomics Tools: RNA-seq analysis, differential gene expression, single-cell RNA-seq (Scanpy, scVI), cell type annotation, transcriptomic data analysis, gene regulatory network analysis, NGS data analysis (Trimmomatic, Bowtie2/STAR), AutoDock, PyRx, GROMACS, Seurat, Discovery Studio, GEO, UCSC Genome Browser, AlphaFold, MODELLER, Galaxy, ExPASy, ProSA-web

Generative AI Tools: Claude, ChatGPT, Copilot, Perplexity (code development, debugging, literature review)

LEADERSHIP & SERVICE

- **Vice President of Operations**, Johns Hopkins Product Management Club, 2026
- **Co-Chair of Events & Programming Committee**, AAP Student Government Association, 2025
- Project development at Kindling by PAVA center, 2025
- **Secretary**, IEEE-Electron Devices Society, 2023
- **Events Head,** Smile Over Stress, 2022